

Blame games: Who is to blame when things go wrong, human or AI?

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Abstract

Understanding responses to AI versus human errors is crucial, as they shape trust, acceptance, and interaction outcomes. This study investigates the emotional dynamics of human-AI interactions by examining how agent identity (human vs. AI) and error severity (low vs. high) influence negative emotional reactions. Using a 2×2 factorial design with 250 participants, the findings reveal that human agents consistently elicit stronger negative emotions than AI agents, regardless of error severity. Moreover, perceived experience moderates this relationship under specific conditions: individuals who view AI less experienced than humans exhibit stronger negative emotions toward human errors, while this effect diminishes when AI is perceived as having higher experience. However, perceived agency does not significantly influence emotional responses. These findings highlight the critical role of agent identity and perceived experience in shaping emotional reactions to errors, providing new insights into the psychological dynamics of human-AI interactions.

CCS Concepts

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

Keywords

Human-AI interaction, Negative emotions, Agent identity, Mind perception

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1 INTRODUCTION

When AI and humans make identical mistakes, people's responses and perceptions often differ significantly [1, 4]. These differences are shaped by factors such as the severity of the mistake, expectations of accountability, and the perceived role of AI in decision-making. Research has shown that AI is often blamed more than

humans for similar errors, especially when the mistake is serious or occurs in high-stakes situations [4, 7]. This tendency may arise from heightened expectations of AI's precision and reliability, as well as the belief that its errors are preventable and indicative of flaws in design or programming [17]. However, when human oversight is involved, blame can shift partially back to the individuals responsible, as their failure to intervene is seen as a contributing factor [2, 18, 27]. The severity of the mistake further amplifies this dynamic, as major errors tend to provoke greater scrutiny of AI systems compared to human faults [7].

Prior research has examined blame attribution across agent identities (human or AI), producing mixed findings. Some studies report that humans are held more accountable for errors due to their perceived autonomy [2, 18]. Conversely, other studies discover that AI is more frequently blamed for mistakes, either because of higher expectations of AI accuracy [17] or the lack of transparency in AI systems [5]. Less is known about the role of individual differences in shaping emotional responses to the same errors made by AI or humans.

To address these gaps, this study will examine the impact of agent identity on negative emotions and explore how individual differences in the perceived agency and experience of AI and humans moderate these emotional responses. Drawing on mind perception theory, we specifically investigate two specific individual differences: perceptions of AI's agency (the ability to act intentionally) relative to humans and perceptions of AI's experience (the ability to feel and sense) relative to humans [8, 9, 31].

2 RELATED WORKS

Understanding responses to identical mistakes made by humans and AI is vital for advancing research on trust, accountability, and decision-making in an increasingly AI-integrated world [1]. Reactions to AI errors often diverge from those directed at human errors due to heightened expectations of AI's precision and reliability, which significantly influence public trust and the adoption of AI technologies [6]. Meanwhile, these differences raise distinct ethical and accountability challenges as AI mistakes are frequently attributed to systemic flaws in design or programming rather than the AI itself, shifting blame to developers or organizations [18, 25].

2.1 Agent identity and responses to errors

A few studies suggest that humans are more likely to be blamed for errors than AI agents. This tendency can be attributed to the perception that humans possess greater agency and intentionality, making their mistakes seem more deliberate or avoidable [13, 27, 33]. For

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example, in driving scenarios, individuals attribute more responsibility to human drivers compared to self-driving cars in the event of accidents, reflecting expectations of human control and accountability [13]. Similarly, when autonomous systems include a manual mode, observers assign more blame to the human operators, as they perceive the option of intervention as increasing human responsibility, even if the operators had no opportunity to act [2]. These findings highlight the significant role of perceived control and agency in shaping blame attribution, consistently making humans more susceptible to blame in error scenarios. Conversely, other researchers find that AI agents often face harsher scrutiny, particularly in high-stakes or critical tasks where optimal performance is expected [4, 7, 14, 19]. For instance, errors made by AI systems are judged more severely because they violate the high expectations of accuracy and reliability associated with technology [17]. In addition, Chen [5] highlights how the “black box” nature of AI decision-making—referring to the lack of transparency in how AI systems make decisions—exacerbates blame. Beyond these factors, AI agents also face harsher judgment in moral dilemmas [21]. These findings demonstrate that the heightened expectations, lack of transparency, and AI-related ethical or moral issues collectively contribute to more severe blame when errors occur.

As AI systems are increasingly integrated into emotionally sensitive and high-stakes domains, such as AI-assisted medical diagnosis [23], understanding the emotional repercussions of their errors becomes not only a theoretical question but also a practical necessity. Thus, the first research question is formulated:

RQ1: Which elicits more intense negative emotions—errors made by humans or errors made by AI agents?

2.2 The role of error severity

Prior research has identified several factors that influence responses to errors during human-AI interactions, such as expectancy violation [24] and task context [13, 20]. These factors are especially pronounced in high-stakes situations, where error severity becomes a critical determinant in shaping emotional responses. Error severity refers to the extent of consequences caused by an error, ranging from low-impact (e.g., errors that do not hinder task completion) to high-impact (e.g., errors that lead to task failure or severe consequences) [15, 30]. The current research aims to explore how error severity may moderate emotional responses to the same errors committed by human versus AI agents. Specifically, high-severity errors tend to elicit stronger negative emotions toward AI agents due to heightened expectations regarding their precision, efficiency, and reliability [5, 17]. These expectations, combined with the perception that AI lacks emotional engagement or the ability to justify its mistakes, may amplify users’ dissatisfaction when severe errors occur. In contrast, users may attribute high-severity errors by human agents to inherent human fallibility, resulting in relatively less severe emotional reactions [21]. In low-severity scenarios, individuals often hold higher emotional expectations for human agents, perceiving them as empathetic and capable of understanding user needs. When human agents fail to meet these expectations, users may feel disappointed, leading to stronger negative emotions compared to AI agents [14, 21]. Based on the works reviewed, two hypotheses are formulated:

H1: In high-severity error scenarios, AI agents could evoke stronger negative emotions than human agents.

H2: In low-severity error scenarios, human agents could evoke stronger negative emotions than AI agents.

2.3 The role of mind perception

Mind perception theory suggests people often attribute varying degrees of mental faculties to non-human entities, including animals, machines, and even inanimate objects [31]. The process of mind perception is based on two primary dimensions: agency and experience [8]. Agency refers to the capacity of an entity to think, plan, and act intentionally. On the other hand, experience refers to the ability to feel emotions, pain, and pleasure. However, individuals’ perception of the minds of other entities is relatively subjective, meaning that different people can have varied mind perceptions of the same entity [8, 34]. This subjectivity can significantly influence emotional responses to errors made by both human and AI agents.

Based on perceived experience, studies have indicated that when entities are perceived as having experience (the ability to feel or sense), they are attributed greater moral responsibility, which influences emotional reactions [10, 31]. For instance, when AI is perceived as having a superior capacity for experience, individuals may expect it to demonstrate more sophisticated emotional responses and understanding, thereby raising the moral bar and intensifying emotional reactions when it fails [32]. Moreover, the way people judge actions is significantly influenced by how much experiential capacity they attribute to an agent. When errors occur, those perceived to have greater experience evoke stronger emotional reactions, reflecting a broader pattern where moral judgments are amplified by perceived capacities to experience harm [11, 26]. Therefore, the following hypothesis is formulated:

H3: Individuals who perceive AI’s experience as superior to that of humans react more negatively to AI errors, whereas those who perceive AI’s experience as inferior react more negatively to human errors.

Furthermore, from the perspective of perceived agency, this dimension focuses on an entity’s capacity for independent thought and decision-making. For example, when an entity is perceived as a moral agent—capable of controlling its actions—individuals tend to expect it to act responsibly. If the entity fails to meet these expectations, it could trigger feelings of moral outrage or disappointment [11]. Moreover, when an entity is viewed as having greater agency, people are more likely to attribute intentionality to its actions, making its mistakes feel more emotionally charged [11]. Therefore, the last hypothesis is proposed:

H4: Individuals who perceive AI as more capable than humans react more negatively to AI errors, whereas those who perceive AI as less capable react more negatively to human errors.

3 METHOD

To examine the hypotheses, we conducted an online experiment ($N = 250$) employing a 2 (agent identity: human vs. AI) $\times 2$ (error severity: low-severity scenario, order errors at a restaurant vs. high-severity scenario, prescription errors at a hospital) factorial design.

3.1 Participants

The experiment was conducted in October 2024 using the Credamo platform, with participants recruited from China. Of the 257 questionnaires collected, 7 were invalid, leaving a 97.3% effective response rate. The sample of 250 has 31.6% male ($n = 79$) and 68.4% female ($n = 171$) participants. Ages ranged from 18 to 64 years, with most of them in the 25–34 age group (45.6%, $n = 114$). Many had a college degree (72.4%).

3.2 Procedure

Participants were randomly assigned to one of the two experimental conditions and presented with a detailed written description of the situation to help them sense the scenarios. In the restaurant scenario, participants imagined ordering food while interacting with a manager and an assistant (human or AI) recording their preferences. In the hospital scenario, they imagined consulting a doctor, assisted by an assistant (human or AI) managing medical records. To ensure the realism of the scenarios, participants rated their perceptions of the scenario's credibility and relatability using a 7-point Likert scale (1 = *strongly disagree*, 7 = *strongly agree*). High ratings ($M = 5.88$, $SD = 0.8$) confirmed the scenarios were perceived as realistic and relatable.

Subsequently, the same mistake was introduced into each of the two scenarios. In the restaurant setting, participants were informed that the assistant had failed to note an explicitly stated dietary restriction, resulting in the inclusion of unwanted ingredients in the meal. In the hospital setting, the assistant mishandled a prescription, including a medication flagged as unsuitable due to the participant's reported allergies. To confirm the hospital error was perceived as more severe, participants rated the error severity on a 5-point Likert scale (1 = *not severe at all* to 5 = *extremely severe*). Afterward, participants completed a survey assessing their negative emotions and perceptions of AI's agency and experience compared to humans.

3.3 Measures

3.3.1 Negative Emotions. Participants' negative emotional responses were measured using an adapted scale [29]. After the error was introduced, they were asked to sense the scenarios and rate their emotions on a 7-point scale anchored by opposing states (e.g., "calm" to "irritated"). Markers placed further right indicated stronger negative emotions. Higher scores reflected greater emotional intensity (Cronbach's $\alpha = .87$, $M = 4.18$, $SD = 1.05$).

3.3.2 Perceived Agency and Experience. Participants' perceptions of AI's cognitive and emotional capabilities relative to humans were assessed through two constructs of mind perception [8]: perceived agency (PA) and perceived experience (PE), adapted from Yam et al. [34]. Perceived agency (Cronbach's $\alpha = .73$, $M = 3.99$, $SD = 1.14$) was measured by assessing participants' beliefs about AI's ability to communicate, think, plan actions, and remember information. Perceived experience (Cronbach's $\alpha = .71$, $M = 2.08$, $SD = 1.03$) was measured by evaluating participants' perceptions of AI's ability to feel pain, fear, desires, and happiness. Participants rated each dimension on a 7-point Likert scale (1 = *far inferior to humans*, 7 = *far superior to humans*). The average scores for PA and PE were

calculated to understand individual differences in how participants evaluated AI's cognitive and emotional capabilities compared to humans.

3.3.3 Scenario Realism. Scenario realism was assessed using a three-item scale adapted from Bagozzi et al. [3]. Participants rated the realism, credibility, and immersiveness of the scenario on a 7-point Likert scale ranging from 1 (*strongly disagree*) to 7 (*strongly agree*), with statements including, "The scenario felt realistic" and "I could immerse myself in the role." (Cronbach's $\alpha = .72$, $M = 5.88$, $SD = 0.81$).

4 RESULTS

4.1 Manipulation check

An independent samples t -test was conducted to compare perceived error severity between the restaurant and hospital scenarios. The manipulation check confirms that errors in the hospital scenario were perceived as significantly more severe ($M = 4.18$, $SD = 0.89$) than those in the restaurant scenario ($M = 2.97$, $SD = 0.96$), $t(248) = 10.38$, $p < .001$. Therefore, the study successfully operationalized low- and high-severity errors through the restaurant and hospital scenarios respectively.

4.2 Agent identity and error severity

A two-way univariate analysis of variance (ANOVA) was conducted to examine the effects of agent identity (Human vs. AI) and error severity (low vs. high) on the intensity of negative emotions elicited after an error. The results revealed a significant main effect of agent identity on negative emotions, $F(1, 246) = 3.948$, $p = .048$. Specifically, participants reported stronger negative emotions when errors were committed by human agents ($M = 4.31$, $SD = 0.97$) compared to AI agents ($M = 4.06$, $SD = 1.11$). Therefore, RQ1 was confirmed, demonstrating that agent identity significantly impacts negative emotional responses.

Our analysis reveals another significant main effect of error severity on negative emotions, $F(1, 246) = 26.805$, $p < .001$. Errors in high-severity contexts (hospital scenario: $M = 4.51$, $SD = 0.87$) elicited significantly stronger negative emotions compared to those in low-severity contexts (restaurant scenario: $M = 3.86$, $SD = 1.10$). However, the interaction effect between agent identity and error severity was not significant, $F(1, 246) = 0.565$, $p = .453$. This indicates that the influence of agent identity on negative emotions does not differ between the levels of error severity. Thus, H1 and H2 are not supported. Regardless of the severity of the error, human agents consistently elicited stronger negative emotions than AI agents.

4.3 The role of perceived experience

To investigate whether participants' perceived experience of AI relative to humans moderates the relationship between agent identity and negative emotions elicited after an error, a General Linear Model (GLM) was employed. The results show that the main effect of agent identity on negative emotions was significant, $F(1, 246) = 4.127$, $p = .043$, with human agents eliciting stronger negative emotions than AI agents. The main effect of perceived experience was also significant, $F(1, 246) = 9.172$, $p = .003$, indicating that

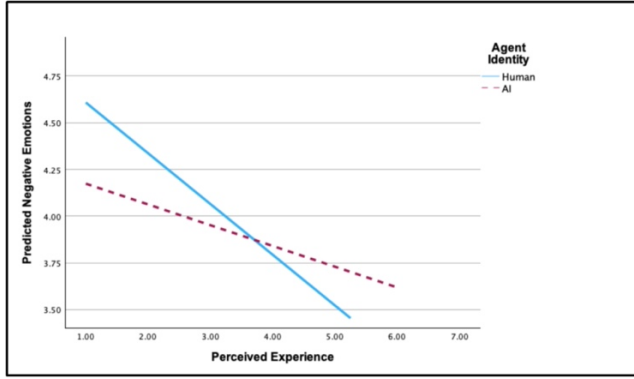


Figure 1: Effects of perceived experience on negative emotions

higher levels of PE were associated with lower negative emotions. However, the interaction effect between agent identity and PE was not significant, $F(1, 246) = 1.613, p = .205$.

To better understand the observed trend, simple main effects analyses were conducted at high ($M + 1SD$) and low ($M - 1SD$) levels of PE. At low PE ($M - 1SD$), there was a significant difference in negative emotions between human agents ($M = 4.59$) and AI agents ($M = 4.17$), $t(246) = 2.328, p = .021$, suggesting that human agents elicited stronger negative emotions than AI agents. This finding supports H3 that individuals who perceive AI's experience as inferior to humans are more likely to experience stronger negative emotions when humans make an error. However, at high PE ($M + 1SD$), the difference in negative emotions between human agents ($M = 4.04$) and AI agents ($M = 3.94$) was not significant, $t(246) = 0.524, p = .600$. This result does not support H3 that individuals who perceive AI's experience as superior to humans would experience stronger negative emotions when AI makes an error. While the findings align with the hypothesis at low levels of PE, the predicted effect at high levels of PE was not observed. This suggests the complex relations between agent identity and negative emotions, meriting further investigation. Thus, H3 was partially supported.

4.4 The role of perceived agency

To examine whether participants' perceived agency of AI relative to humans moderates the relationship between agent identity and negative emotions, a GLM was used. The analysis revealed no significant effects overall. Specifically, the main effect of agent identity was not significant, $F(1, 246) = 0.615, p = .434$, and the main effect of perceived agency was also not significant, $F(1, 246) = 3.766, p = .053$. Finally, the interaction effect between agent identity and perceived agency was not significant, $F(1, 246) = 0.008, p = .930$.

Despite the insignificance in the main analysis, simple main effects were examined at high ($M + 1SD$) and low ($M - 1SD$) levels of PA to explore potential patterns. At low levels of PA ($M - 1SD$), human errors ($M = 4.53$) elicited stronger negative emotions compared to AI errors ($M = 4.13$), but this difference was not statistically significant ($t(246) = 1.854, p = .065$). At high levels of PA ($M + 1SD$), negative emotions elicited by human errors ($M = 4.23$)

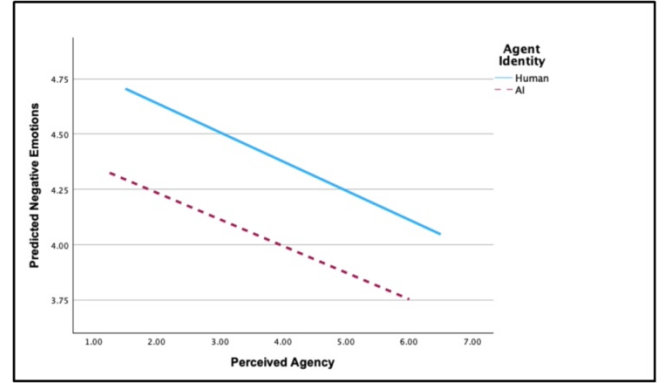


Figure 2: Effects of perceived agency on negative emotions

were similarly higher than those elicited by AI errors ($M = 3.86$), though this difference was also not statistically significant ($t(246) = 0.287, p = .711$). Overall, the results did not support H4, as the interaction effect between agent identity and perceived agency was not statistically significant.

5 DISCUSSION

5.1 Agent identity and error severity

The findings of this study align with existing literature in multiple aspects. Research indicates that humans are often assigned more responsibility than artificial intelligence when making errors in comparable situations. For instance, Arnestad et al. [2] observe that the presence of a manual mode in AI systems significantly increases blame attributed to human operators, even when they do not actively intervene. However, the results of this study offer a different perspective on some prevailing views in the existing literature. Some studies have found that AI errors, particularly in specific contexts such as high-risk tasks or technical failures, can lead to more severe negative consequences compared to those caused by humans. For example, the severity of scrutiny on AI errors stems from these high expectations [17]. However, these studies mainly focus on blame attribution rather than on emotional responses, which is the foci of our study.

Our research finds that human errors consistently trigger stronger negative emotional reactions than AI errors across various contexts, regardless of the severity of the error. This finding can be explained by higher expectations of human emotion and moral responsibility. When people perceive an entity as possessing emotions, they are more likely to exhibit emotional reactions to its behavior. Conversely, if people perceive an entity as lacking emotions, their emotional investment in its actions tends to be lower, resulting in less intense reactions [11, 26, 35]. Specifically, individuals may perceive human errors as having stronger emotional motivations and moral responsibilities, leading to more intense emotional reactions. In contrast, AI is perceived as lacking emotional experience and moral responsibility [27, 33], so its mistakes carry less emotional weight and lead to weaker emotional reactions.

5.2 Perceived experience and agency

Regarding the moderating role of perceived experience, our results offer a different perspective from Yam et al. [34], who demonstrated that high levels of perceived experience in service robots can attenuate negative reactions to service failures, as customers may attribute humanlike qualities such as guilt and remorse to the robots, thereby fostering empathy and reducing dissatisfaction. However, our results are consistent with research suggesting that when an entity is perceived to have stronger experience capabilities, individuals raise their moral standards for it and intensify their emotional reactions to its failures [11, 26, 32]. Our results reveal this effect under specific conditions. Specifically, at low levels of perceived experience (where AI is perceived as inferior to humans in emotional capabilities), human agents elicited significantly stronger negative emotions than AI agents. However, at high levels of perceived experience (where AI is perceived as superior to humans), the emotional response difference between human and AI agents diminished. This suggests that under high PE conditions, the perceived emotional responsibility between human and AI agents becomes less distinct, leading to more similar emotional reactions to their errors.

In terms of perceived agency, this study did not observe a significant interaction effect between agent identity and perceived agency. While prior research has highlighted the role of perceived agency in influencing error attribution to entities (e.g., [2, 18, 25]), our findings suggest that perceived agency does not significantly moderate emotional reactions to errors. One possible explanation is that individuals' emotional responses in this study's context were more influenced by perceived experience than by perceived agency. Specifically, people's reactions to errors made by human and AI agents may be shaped more by their expectations of the agents' emotional capacities, at least within the cultural context examined, rather than by their perceptions of control or decision-making capabilities. This study provides empirical evidence supporting the role of perceived experience in shaping emotional responses, though further research is needed to explore its relative importance across different contexts.

Notably, in our sample, the average perceived experience rating for AI relative to humans was 2.08, indicating that most participants perceived AI's emotional experience to be much lower than that of humans. This finding aligns with the widespread belief that AI lacks emotion [28], further supporting our interpretation of the emotional response differences between human and AI errors. Human emotional experiences are generally considered more complex and richer, which likely explains why people react more strongly to human errors. Human errors are often perceived as influenced by emotions and associated with greater moral responsibility, whereas AI is regarded as an emotionless tool or system. Consequently, errors made by AI are typically seen as technical issues, leading to weaker emotional responses. Therefore, this result suggests that differences in perceived experience may be a key factor influencing emotional responses. Individuals are more likely to attribute greater emotional responsibility and moral evaluation to human errors, which leads to stronger emotional reactions. In contrast, AI errors, perceived as less emotionally complex, are often

regarded as more objective or systemic issues, thereby evoking weaker emotional responses.

6 Limitations and Future Directions

Despite some insights produced, several limitations should be acknowledged. First, the role-play online experiment, though validated in service failure scenarios [16, 18], cannot fully replicate real-life emotional responses. Although participants rated scenario realism highly ($M = 5.88$ out of 7), their emotional reactions may differ from those in actual situations, a limitation inherent to role-play studies [12]. Second, the sample consisted entirely of participants from China, which introduces potential cultural specificities that may limit the generalizability of the findings. For example, in healthcare contexts, Chinese participants may exhibit unique emotional responses due to cultural attitudes toward medical care, such as higher sensitivity to medical errors or a greater emphasis on hierarchical relationships between doctors and patients [22]. These cultural factors might have influenced participants' perceptions and emotional reactions, highlighting the need for caution when extending the findings to other cultural contexts. Third, although the value of this study lies in revealing that in different agent identities, humans elicit stronger negative emotions than AI in response to the same error, and in highlighting the moderating role of individual perception differences regarding mind perception of AI and humans, the study did not explore the detailed mechanisms underlying these effects.

Future research should address these limitations by incorporating more immersive experimental designs, such as virtual reality or real-world scenarios, to better simulate actual experiences and elicit more authentic emotional responses. However, it is equally important to consider the ethical implications of inducing negative emotions in real-life scenarios, ensuring that participants' well-being is safeguarded. Finally, future studies could focus on exploring the psychological mechanisms driving differences in emotional reactions to AI and human agents. One approach could be investigating the role of attribution theory—whether individuals attribute errors made by humans to controllable factors (e.g., negligence) and AI errors to uncontrollable factors (e.g., system failure), leading to stronger negative emotions toward human agents.

7 Conclusion

This study provides insights into the emotional dynamics of human-AI interactions, particularly in error scenarios. We find that human agents consistently evoke stronger negative emotional responses than AI agents, regardless of error severity, reflecting higher expectations of human emotional capacity and moral responsibility. While perceived experience influenced overall emotional responses, it did not consistently moderate the ties between agent identity and negative emotions. Similarly, perceived agency showed no significant effect, underscoring the dominance of agent identity over individual differences in shaping emotional reactions to errors. These results highlight the importance of considering emotional responses as a critical dimension in human-AI interaction research, as they offer unique insights into how individuals evaluate different agents in error scenarios. Understanding these emotional reactions is crucial not only for advancing theoretical frameworks in

human-AI interaction but also for designing more effective and empathetic AI systems that can better align with human expectations and emotional needs.

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